

Artifact Removal for ICMS-Evoked LFP

Recordings

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by

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Declaration of Originality

I, **Ramya Sree Galanki**, with SR No. **04-02-02-10-51-24-1-24770** hereby declare that the material presented in the thesis titled

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AI-assisted writing tools, including ChatGPT and Grammarly, were used during the preparation of this report for grammar correction, language refinement, code-debugging guidance, and improving clarity of explanation. All research design, implementation, analysis, results, interpretations, and conclusions presented in this report are my own work.

Abstract

Intracortical microstimulation (ICMS) is an important technique for studying and modulating neural circuits, but electrical stimulation introduces large artifacts into simultaneously recorded local field potential (LFP) signals. These artifacts can be much larger than the neural activity of interest and can obscure the early post-stimulation response. The aim of this project was to develop and evaluate artifact-removal methods for ICMS-evoked LFP recordings from a microelectrode array experiment.

Several artifact-removal approaches were implemented and compared, including ERP-based template subtraction, aligned and pulse-wise template subtraction, PCA-based artifact removal, SMARTALite nearest-neighbor template subtraction, and SMARTAFull with optimal shrinkage. Initial methods showed that a single fixed template was not sufficient because the artifact varied across trials, pulses, and electrodes. PCA-based methods reduced stimulation harmonics strongly, but they were treated cautiously because aggressive component removal can also subtract time-locked neural activity.

A SMARTA-inspired approach was then developed. SMARTALite used local nearest-neighbor pulse templates, while SMARTAFull additionally used high-pass similarity features and optimal-shrinkage denoising before selecting similar artifact segments. After enabling the optimal-shrinkage dependency, SMARTAFull with $K = 3$ and a 100 Hz high-pass similarity filter became the final selected method. In the final real-data validation across 31 selected electrodes, SMARTAFull achieved the lowest mean total FFT error, lowest mean stimulation-harmonic excess error, and lowest balanced score among the tested methods. It was the best method on 30 of 31 electrodes for total FFT error and on all 31 electrodes for harmonic artifact error. ERPSubtraction remained the most conservative negative-control method, but SMARTAFull provided the strongest overall artifact reduction and the best trade-off among the evaluated methods.

This work presents a practical artifact-removal pipeline for ICMS-evoked LFP recordings and provides a basis for further analysis of stimulation-evoked neural activity.

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Chapter 1

Introduction

1.1 Background and Motivation

Electrical stimulation is widely used in neuroscience to study how neural circuits respond to controlled external input. In intracortical microstimulation (ICMS), short electrical pulses are delivered through microelectrodes placed within the cortex. This makes it possible to activate local neural populations and study the resulting activity in nearby or connected brain regions. ICMS is useful both as an experimental tool for understanding neural function and as a possible approach for neural prosthetic and neuromodulation applications.

In many experiments, stimulation is applied while neural signals are recorded from the same or nearby electrode array. Local field potentials (LFPs) are commonly used in such studies because they reflect the summed activity of local neural populations over a relatively low-frequency range. However, simultaneous stimulation and recording introduces a major signal-processing challenge. The stimulation pulse produces a large electrical artifact in the recorded signal. This artifact can be much larger than the neural response of interest and may overlap with the time period in which the evoked neural activity is expected to occur.

The artifact is difficult to remove because it is not always identical across trials, pulses, and electrodes. A simple average template may remove the dominant artifact shape, but it may not handle smaller variations in amplitude, timing, or waveform shape. More flexible methods, such as aligned template subtraction, PCA-based subtraction, and pulse-wise template estimation, can address some of these variations, but they must be designed carefully to avoid removing neural

signal along with the artifact.

The motivation for this project is to develop and evaluate artifact-removal methods for ICMS-evoked LFP recordings. The goal is not only to reduce the visible stimulation artifact, but also to preserve the underlying LFP structure so that the cleaned signal can be used for further analysis of stimulation-evoked neural activity.

1.2 Problem Statement

Intracortical microstimulation produces large electrical artifacts in recorded LFP signals. These artifacts occur at stimulation times and can extend into the early post-stimulation period, where neural responses may also be present. As a result, the recorded signal contains a mixture of stimulation artifact and neural activity, making direct interpretation difficult.

A major challenge is that the artifact is not perfectly constant. Its shape and amplitude can vary across trials, pulses, and electrodes. Therefore, a single fixed artifact template may not be sufficient for reliable removal. At the same time, more aggressive methods may remove parts of the neural signal along with the artifact. This creates a trade-off between artifact suppression and signal preservation.

The problem addressed in this work is to remove stimulation artifacts from ICMS-evoked LFP recordings while preserving the underlying neural signal as much as possible. The cleaned signal should be closer to the no-stimulation reference in time-domain and frequency-domain characteristics, without introducing artificial distortions or excessive smoothing across trials.

1.3 Objectives

The main objective of this project is to develop and evaluate artifact-removal methods for ICMS-evoked LFP recordings. The specific objectives are:

- To study the structure of stimulation artifacts in LFP recordings obtained during ICMS experiments.
- To implement and compare ERP-based, PCA-based, SMARTALite, and SMARTAFull artifact-removal methods.

- To evaluate artifact suppression using FFT-based error, stimulation-harmonic error, and pulse-triggered residual metrics.
- To evaluate signal preservation using no-stimulation negative-control distortion, trial-to-trial variability, and semi-synthetic recovery.
- To identify a practical artifact-removal pipeline that reduces stimulation artifacts while preserving the overall structure of the LFP signal.

1.4 Contribution

The main contribution of this work is the development and evaluation of an artifact-removal pipeline for ICMS-evoked LFP recordings. Several methods were implemented and compared, including ERP-based template subtraction, PCA-based subtraction, SMARTALite nearest-neighbor template subtraction, and SMARTAFull with optimal shrinkage.

A key contribution is the adaptation of SMARTA-inspired similarity-based template estimation to short-train ICMS artifacts. Instead of estimating a single global artifact template, the SMARTA-based methods estimate local templates from similar pulse segments. This improves flexibility when artifacts vary across trials, pulses, and electrodes.

During development, SMARTALite ensemble with $K = 3$ and $K = 5$ provided a useful lightweight adaptive method. After resolving the optimal-shrinkage dependency, SMARTAFull with $K = 3$ and a 100 Hz high-pass similarity filter was evaluated across the final 31 selected V1 electrodes. This method produced the lowest mean total FFT error, lowest mean harmonic artifact error, and lowest balanced score. Therefore, SMARTAFull was selected as the final method for this dataset.

The work also includes systematic validation using real high-stimulation trials, no-stimulation negative controls, pulse-triggered residual analysis, and semi-synthetic validation. Together, these analyses quantify both artifact reduction and the risk of over-cleaning.

The dataset and preliminary GUI/common data-access programs were provided by the NeuroOscillations Laboratory, while the artifact-removal comparisons, SMARTALite/SMARTAFull validation pipeline, metric design, and final across-electrode evaluation were carried out as part of this project.

1.5 Outline of Report

Chapter 1 introduces the background, motivation, problem statement, objectives, and contributions of the work. Chapter 2 presents the literature survey on LFPs, ICMS artifacts, conventional artifact-removal methods, PCA-based methods, and SMARTA. Chapter 3 describes the dataset, preprocessing, artifact-removal methods, and evaluation metrics. Chapter 4 presents experimental results, including ERP, PCA, SMARTALite, SMARTAFull, across-electrode validation, and preservation checks. Chapter 5 concludes the report and discusses possible future work.

Chapter 2

Literature Survey

2.1 Local Field Potentials and Neural Oscillations

Neural activity can be recorded at different spatial and temporal scales, ranging from single-unit spiking activity to macroscopic signals such as EEG and ECoG. Local field potentials (LFPs) are recorded using microelectrodes placed within brain tissue and mainly reflect the summed electrical activity of a local neural population. Compared with single-unit activity, LFPs provide a broader view of population-level activity and are therefore useful for studying coordinated neural responses.

Brain activity contains oscillations over a wide range of frequencies, including low-frequency rhythms as well as beta and gamma-band activity. These oscillations are thought to reflect temporal organization of neural populations. Buzsáki emphasized that neuronal oscillations provide an important timing structure for communication and coordination within and across brain networks [1].

In visual cortex, LFP recordings are commonly used to study stimulus-induced responses. Visual stimuli such as gratings can produce gamma-band oscillations, typically in the range of approximately 30–80 Hz. These responses depend on stimulus properties such as size, contrast, orientation, spatial frequency, temporal frequency, and chromatic content [2, 3]. Therefore, artifact-removal methods must preserve both time-domain waveform structure and frequency-domain components that may reflect meaningful neural activity.

2.2 Intracortical Microstimulation and Stimulation Artifacts

Intracortical microstimulation is used to activate neural tissue by delivering small electrical pulses through microelectrodes implanted in the cortex. It is widely used to study causal relationships between neural activity and behavior and is also relevant for cortical prostheses and neuromodulation systems [10].

A major difficulty in simultaneous stimulation and recording is the presence of stimulation artifacts. These artifacts arise from the stimulation pulse, electrode-tissue interface, and recording hardware. They can be much larger than the neural signal and may obscure the early post-stimulation response. The artifact is usually strongest near the stimulation electrode and may affect both time-domain waveforms and frequency-domain spectra.

Repeated stimulation pulses introduce strong spectral components at the stimulation frequency and its harmonics. If these components are not removed properly, they can be mistaken for neural oscillations or can mask genuine neural activity. This motivates artifact-removal methods that reduce pulse-locked contamination while preserving neural structure.

2.3 Conventional Artifact-Removal Methods

Conventional artifact-removal methods include template subtraction, interpolation, curve fitting, filtering, and moving-average template methods. Template subtraction estimates an average artifact waveform and subtracts it from each contaminated signal. It is simple and interpretable, but it assumes that the artifact shape is stable. If the artifact varies across trials or pulses, residual artifacts may remain.

Interpolation-based methods remove or replace artifact-contaminated samples around the stimulation pulse. These methods can be useful when the artifact is short, but they may distort the signal when meaningful neural activity occurs close to the stimulation time. Curve fitting models the artifact waveform using polynomial or exponential functions, but performance depends on how well the model matches the true artifact.

Moving-average subtraction is a more adaptive template method that estimates local templates using nearby artifacts. Sun and Hinrichs proposed a moving-average subtraction method for removing DBS artifacts from EEG and LFP recordings [8]. Such methods motivate adaptive approaches

that estimate local artifact templates rather than using a single fixed waveform.

2.4 Subspace and PCA-Based Artifact Removal

Subspace and decomposition-based methods remove artifacts by representing the recorded signal using components or basis functions. Principal component analysis (PCA), independent component analysis, artifact subspace reconstruction, and SVD-based denoising belong to this general class.

PCA-based methods are attractive because stimulation artifacts often contribute large variance and may be captured by a small number of principal components. Removing those components can suppress artifact-related structure. However, PCA can also be aggressive. If neural activity is time-locked to stimulation or shares variance with the artifact, PCA components may contain both artifact and neural signal. Removing too many components may therefore reduce the visible artifact but also suppress neural activity. This issue motivated the use of negative-control and preservation metrics in the present work.

2.5 SMARTA and Similarity-Based Template Estimation

Similarity-based template estimation is a more adaptive approach to stimulation artifact removal. Instead of assuming that all artifacts can be represented by one average template, these methods estimate an artifact template for each stimulation event using other artifact segments with similar waveform shape.

SMARTA, or Shrinkage and Manifold-based Artifact Removal using Template Adaptation, was proposed for removing stimulation artifacts from LFP recordings. It segments artifacts, denoises artifact segments, identifies similar artifacts using a distance measure, and subtracts a median template estimated from the selected neighbors [12]. A key component of SMARTA is optimal shrinkage for matrix denoising, which improves similarity estimation by reducing noise in the artifact-segment matrix [13].

In this project, SMARTA was adapted to ICMS-evoked LFP recordings. SMARTALite used local nearest-neighbor pulse templates without optimal shrinkage. SMARTAFull used a high-pass similarity representation and optimal-shrinkage denoising before nearest-neighbor selection. After enabling the required optimal-shrinkage helper functions, SMARTAFull became the strongest final

method.

2.6 Summary

The literature shows that LFP recordings provide useful information about population-level neural activity, but simultaneous electrical stimulation introduces large artifacts. Conventional template methods are interpretable but limited when artifact shape varies. PCA-based methods are powerful but can be aggressive and may remove neural signal. SMARTA-style methods provide an adaptive alternative by estimating local templates from similar artifact segments. This motivates the evaluation of SMARTALite and SMARTAFull for ICMS-evoked LFP artifact removal.

Chapter 3

Problem Formulation and Methodology

3.1 Overview

This chapter describes the methodology used for stimulation artifact removal in ICMS-evoked LFP recordings. The overall goal was to reduce electrical stimulation artifacts while preserving the underlying neural signal. Since the true artifact-free neural response is not directly available in real stimulation trials, the methods were evaluated using both real-data comparisons and semi-synthetic validation.

The methodology consisted of four stages. First, LFP data were loaded and no-stimulation and high-stimulation conditions were selected. Second, several artifact-removal methods were implemented, including ERPSubtraction, PCATemplate, SMARTALite, and SMARTAFull. Third, the methods were evaluated using time-domain and frequency-domain metrics. Finally, the selected method was validated across 31 non-stimulating V1 recording electrodes and using additional preservation checks.

3.2 Dataset and Experimental Conditions

The data consisted of LFP recordings from a microelectrode array experiment during visual stimulation and ICMS. The analysis was restricted to the GRF_001 protocol, an amplitude-varying stimulation protocol. In this protocol, the stimulation amplitude values were 0, 2, 4, 8, 16, 32, and 64 μA .

The LFP recordings were stored separately for each electrode. Each electrode file contained an `analogData` matrix with trials as rows and time samples as columns:

$$\text{analogData} \in \mathbb{R}^{N_{\text{trials}} \times N_{\text{time samples}}}.$$

The time axis was stored in `timeVals`. The sampling interval was 0.0005 s, corresponding to a sampling frequency of 2000 Hz.

Two main conditions were used for artifact-removal evaluation. The no-stimulation condition was indexed as $\{1, 1, 1, 5, 5, 4\}$ and contained 192 trials. The high-stimulation condition was indexed as $\{7, 1, 1, 5, 5, 4\}$ and also contained 192 trials. The high-stimulation condition corresponded to 64 μA and was treated as the contaminated condition for the main comparison.

The main artifact-removal and FFT analysis window was 0–0.4 s after stimulation onset. With a 2000 Hz sampling frequency, this window contained 800 samples. The main frequency range for quantitative error metrics was 0–200 Hz.

Table 3.1 Summary of the GRF_001 LFP data used in this project.

Item	Value
Signal type	Local field potential (LFP)
Protocol analyzed	GRF_001
Protocol type	Amplitude-varying stimulation protocol
Sampling interval	0.0005 s
Sampling frequency	2000 Hz
No-stimulation condition	$\{1, 1, 1, 5, 5, 4\}$
High-stimulation condition	$\{7, 1, 1, 5, 5, 4\}$
No-stimulation trials	192
High-stimulation trials	192
Main artifact/FFT window	0–0.4 s
FFT samples	800
Main frequency range	0–200 Hz

The experimental data were recorded in the NeuroOscillations Laboratory (NeO Lab), Centre for Neuroscience, Indian Institute of Science. The preliminary GUI-based analysis framework and common data-loading routines were also developed in the NeO Lab and were used as the starting point for accessing and organizing the dataset. The artifact-removal methods and validation analyses described in this report were implemented and evaluated as part of the present project.

3.3 Preprocessing and Electrode Selection

Electrode selection followed the logic used in the existing ICMS analysis GUI. Visually responsive high-RMS electrodes were loaded from the RFData file and restricted to electrodes numbered 1–48, corresponding to the V1 subset. Bad channels, if identified from impedance measurements, were excluded.

Using this procedure, 32 good electrodes were identified before excluding the stimulation electrode. Electrode 1 was used for stimulation and was therefore excluded from the across-electrode artifact-removal analysis. The final validation used 31 recording electrodes:

3, 4, 5, 6, 7, 8, 10, 11, 13, 14, 15, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 34, 35, 36, 41, 42.

3.4 Problem Formulation

For a high-stimulation trial, the recorded LFP can be modeled as

$$x(t) = s(t) + a(t) + \eta(t),$$

where $x(t)$ is the observed contaminated signal, $s(t)$ is the neural signal, $a(t)$ is the stimulation artifact, and $\eta(t)$ represents noise and trial-to-trial variability. Artifact removal estimates $\hat{a}(t)$ and computes

$$\hat{s}(t) = x(t) - \hat{a}(t).$$

The objective is to reduce $a(t)$ while preserving $s(t)$. In real stimulation trials, $s(t)$ is unknown, so the cleaned high-stimulation signal was compared with a no-stimulation reference. In semi-synthetic validation, the clean signal was known before artifact addition.

3.4.1 Illustration of the Artifact-Removal Goal

The stimulation artifact can be seen clearly by comparing no-stimulation and high-stimulation trials. In the high-stimulation condition, the artifact appears as large pulse-locked deflections during the stimulation period. These deflections are much larger than the background LFP activity and can dominate both the time-domain waveform and the frequency-domain spectrum.

Figure 3.1 shows artifact-contaminated high-stimulation LFP trials from the stimulation electrode, a nearby analyzed electrode, and a distant analyzed electrode. The artifact is pulse-locked across trials, but its amplitude and waveform shape vary across electrodes. This motivates the need for artifact-removal methods that are evaluated across multiple electrodes rather than only at a single channel.

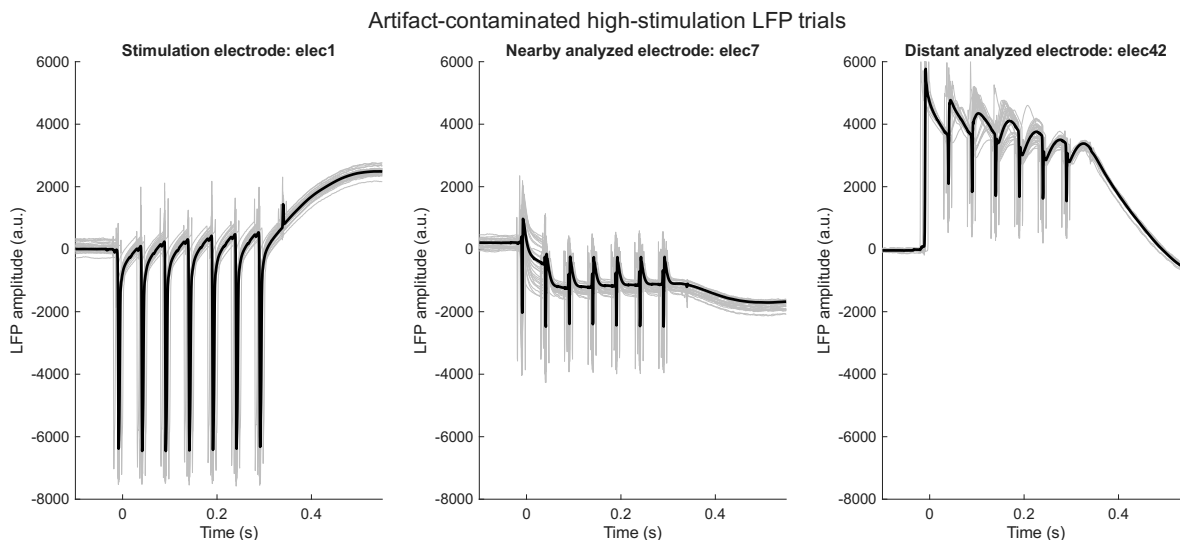


Fig. 3.1 Example artifact-contaminated high-stimulation LFP trials before artifact removal. The stimulation electrode shows large pulse-locked artifacts. Nearby and distant analyzed electrodes also show stimulation-related contamination, but with different amplitudes, polarities, and waveform structure. Gray traces show a subset of high-stimulation trials, and the black trace shows the mean across high-stimulation trials. Electrode 1 was used to illustrate the artifact strength and was excluded from the across-electrode method validation.

Figure 3.2 shows the time-domain difference between the no-stimulation reference condition and the high-stimulation condition for an example analyzed electrode. The no-stimulation trials represent the LFP response in the absence of electrical stimulation, while the high-stimulation trials show additional stimulation-related contamination.

3.4.2 Frequency-Domain Effect of Stimulation

In addition to the time-domain artifact, stimulation also affects the frequency-domain structure of the LFP signal. The FFT was therefore used to compare the no-stimulation and high-stimulation conditions before artifact removal. This comparison was used to show the spectral effect of stimulation and to motivate the FFT-based evaluation metrics used later in this work.

Figure 3.3 shows the FFT comparison for the stimulation electrode and for an example analyzed

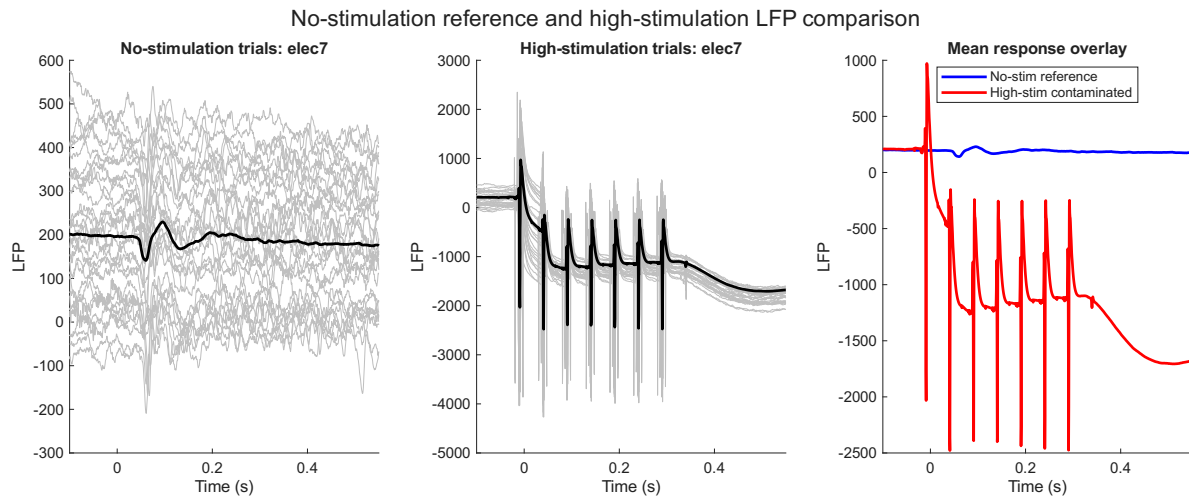


Fig. 3.2 Time-domain comparison between no-stimulation and high-stimulation LFP trials for an example analyzed electrode. The no-stimulation condition is used as the reference signal structure, while the high-stimulation condition contains stimulation-related contamination. Gray traces show individual trials and the black traces show the mean response across trials.

electrode. The stimulation electrode is shown because it clearly illustrates the strength of the artifact, while the example analyzed electrode shows that stimulation-related spectral contamination is also present in a channel used for analysis.

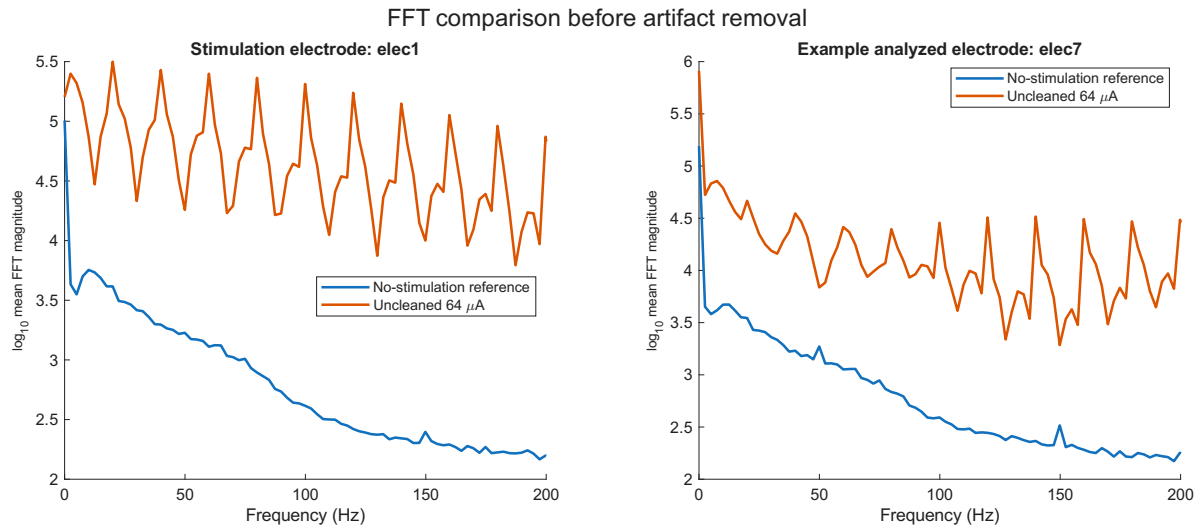


Fig. 3.3 Frequency-domain comparison before artifact removal for the stimulation electrode and an example analyzed electrode. The FFT was computed over the 0–0.4 s analysis window, and the mean log-magnitude spectrum was calculated across trials. Electrode 1 shows strong stimulation-related spectral contamination because it was the stimulation electrode. Electrode 7 shows that stimulation-related contamination is also present in an analyzed electrode.

3.5 Artifact-Removal Methods

3.5.1 ERPSubtraction

ERPSubtraction estimates the average stimulation-locked waveform from high-stimulation trials and subtracts it from each contaminated trial within the artifact window. It is simple and conservative, but it can remove time-locked neural activity if such activity is included in the average template.

3.5.2 PCATemplate

PCATemplate is a decomposition-based baseline. The method models artifact-related variation using principal components estimated within the artifact window. In this project, PCATemplate with $K = 10$ components was used as the PCA comparison method. PCA can reduce harmonic artifacts strongly, but it may be aggressive because artifact-related principal components can contain neural activity as well.

3.5.3 SMARTALite Ensemble

SMARTALite is a simplified SMARTA-inspired method. It segments the stimulation artifact around each pulse, finds nearest-neighbor pulse segments, estimates a local median template, and subtracts that template from the target segment. SMARTALite ensemble K3K5 averages the cleaned outputs obtained using $K = 3$ and $K = 5$. This method served as a lightweight adaptive baseline.

3.5.4 SMARTAFull with Optimal Shrinkage

SMARTAFull extends SMARTALite by improving similarity estimation. Pulse segments are first represented using a high-pass similarity feature, and optimal-shrinkage denoising is applied to the artifact-segment matrix. Nearest neighbors are then selected using the denoised representation, and a median template is estimated from similar segments.

The high-pass filtered representation was used only for similarity estimation and nearest-neighbor selection. The artifact template was subtracted from the original LFP signal, so the final cleaned signal was not simply a high-pass filtered version of the data.

For each pulse-centered artifact segment, SMARTAFull formed a segment matrix across trials and pulses. A high-pass filtered version of the segment matrix was used only for similarity es-

timation. Optimal-shrinkage denoising was then applied to improve the representation used for nearest-neighbor selection. For each target segment, the K nearest denoised segments were selected, and the median of the corresponding original LFP segments was used as the artifact template. This template was subtracted from the original unfiltered LFP segment.

In the final validation, SMARTAFull used $K = 3$, a 100 Hz high-pass similarity filter, and optimal-shrinkage denoising. The active implementation used the project version of `SMARTAFull.m` and `optimal_shrinkage_color_fast.m`, with the helper function `createPseudoNoise.m` added from the original SMARTA repository. The final validation confirmed that the optimal-shrinkage routine was used successfully for SMARTAFull.

3.6 Evaluation Metrics

3.6.1 Frequency-Domain Error

The mean log-magnitude FFT spectrum was computed over the 0–0.4 s window. If $F_{\text{nostim}}(f)$ is the no-stimulation spectrum and $F_{\text{clean}}(f)$ is the cleaned high-stimulation spectrum, the total FFT error was

$$E_{\text{FFT}} = \|F_{\text{clean}}(f) - F_{\text{nostim}}(f)\|_{0-200 \text{ Hz}}.$$

3.6.2 Harmonic Artifact Error

Stimulation-related harmonic error was computed at the stimulation frequency and its harmonics up to 200 Hz. The harmonic excess error measured the positive excess of the cleaned spectrum relative to the no-stimulation reference at these harmonic frequencies. This metric directly assessed residual stimulation-locked spectral contamination.

3.6.3 Negative-Control Distortion

To test over-cleaning, each method was also applied to no-stimulation data. Since no stimulation artifact is present in this condition, a safe method should not strongly change the signal. Negative-control distortion was measured using RMS distortion, normalized RMS distortion, FFT distortion, and trial-variability distortion.

3.6.4 Trial-to-Trial Variability

Artifact removal should preserve realistic trial-to-trial variability. The standard deviation across trials was compared before and after cleaning. Methods that make trials too similar may remove meaningful neural variability along with the artifact.

3.6.5 Balanced Score

A normalized balanced score was computed by combining multiple metrics, including total FFT error, harmonic excess error, below-reference error, trial-variability error, no-stimulation RMS distortion, and no-stimulation FFT distortion. Lower balanced score indicates a better overall trade-off between artifact suppression and signal preservation.

3.6.6 Best-Method Count

For each metric and electrode, the method with the lowest error was identified. The number of electrodes won by each method was counted. This provided a robustness measure across the 31 selected electrodes.

Chapter 4

Experimental Validation

4.1 Overview

This chapter presents the experimental validation of the artifact-removal methods. The methods were first studied using representative examples and then evaluated across the 31 selected non-stimulating V1 recording electrodes. The main comparison used the 0–0.4 s analysis window and the 0–200 Hz frequency range.

4.2 ERP-Based Baseline Results

ERP-based methods were evaluated as interpretable baselines. These included ERPSubtraction, shifted ERP subtraction, scaled ERP subtraction, aligned ERP subtraction, pulse-wise template subtraction, and aligned pulse-wise subtraction.

Table 4.1 FFT-error improvement for ERP-based methods on the stimulation electrode and an example non-stimulating analyzed electrode.

Method	elec1 improvement (%)	elec7 improvement (%)
ERPSubtraction	22.27	25.58
ERPShifted	29.95	30.50
ERPScaled	22.87	26.11
ERPShiftedScaled	29.55	17.94
ERPAligned	31.30	30.07
PulsewiseTemplate	17.83	2.06
ERPAlignedPulsewise	31.62	30.80

ERPShifted, ERPAigned, and ERPAignedPulsewise were the strongest ERP-based variants. ERPSubtraction remained useful as the simplest conservative baseline. However, the ERP methods did not completely remove stimulation-related spectral structure, motivating more adaptive methods.

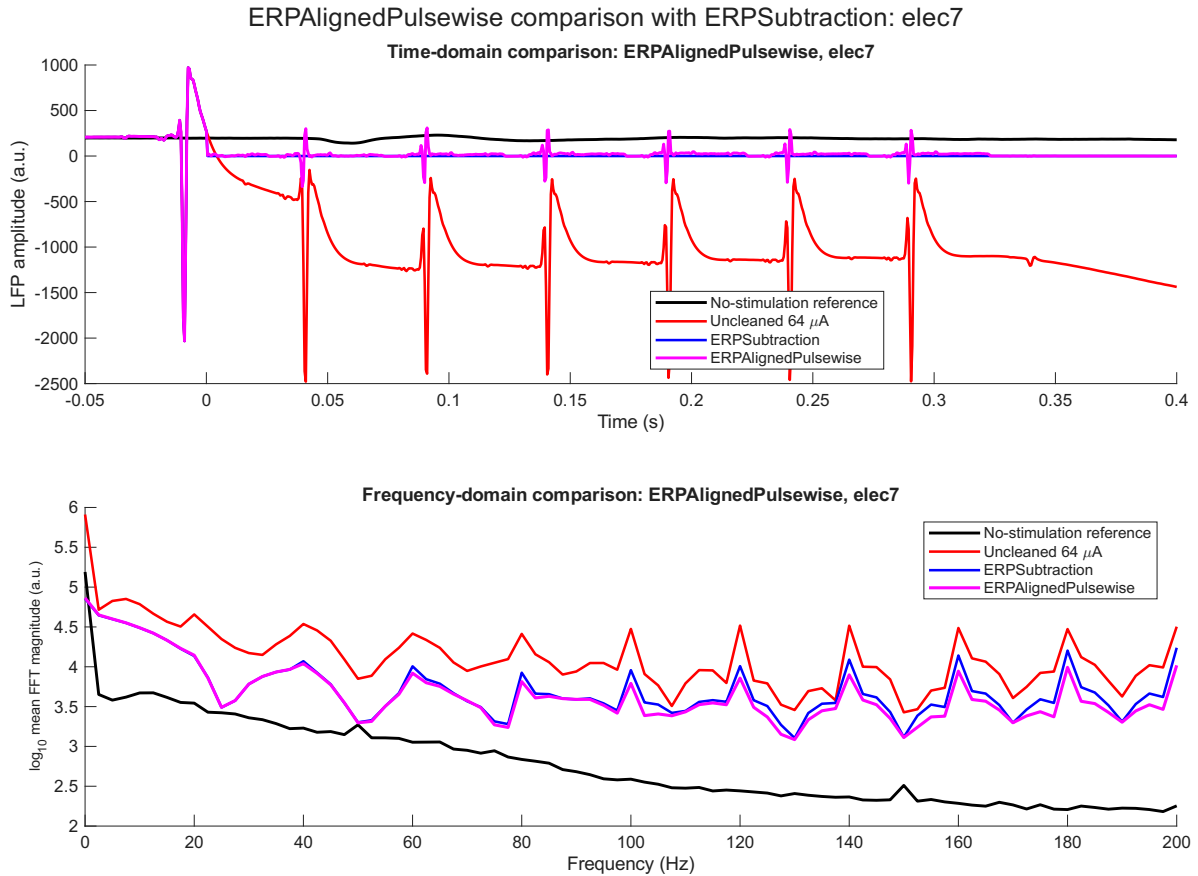


Fig. 4.1 Comparison of simple ERPSubtraction and ERPAignedPulsewise for an example analyzed electrode. The spectra were computed over the 0–0.4 s artifact-analysis window. ERPSubtraction reduced the stimulation artifact relative to the uncleaned high-stimulation signal, while ERPAignedPulsewise gave a small additional reduction.

4.3 PCA-Based Results

PCA-based artifact removal was evaluated as a decomposition-based baseline. The number of PCA components controls the aggressiveness of the method. Larger values can reduce artifact-related spectral structure but may also remove neural signal.

Cross-validation showed that performance improved as K increased from 0 to 3 and then saturated. For the final PCA comparison, $K = 10$ was used.

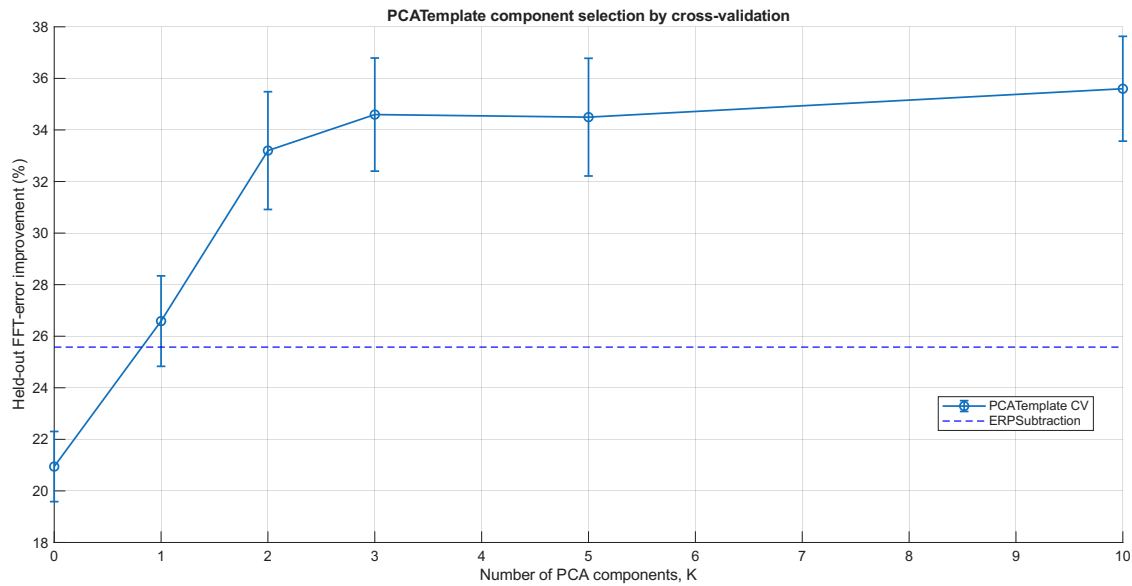


Fig. 4.2 Cross-validation of the number of PCA components used in PCATemplate artifact removal for an example analyzed electrode. Performance improved as the number of PCA components increased and then saturated.

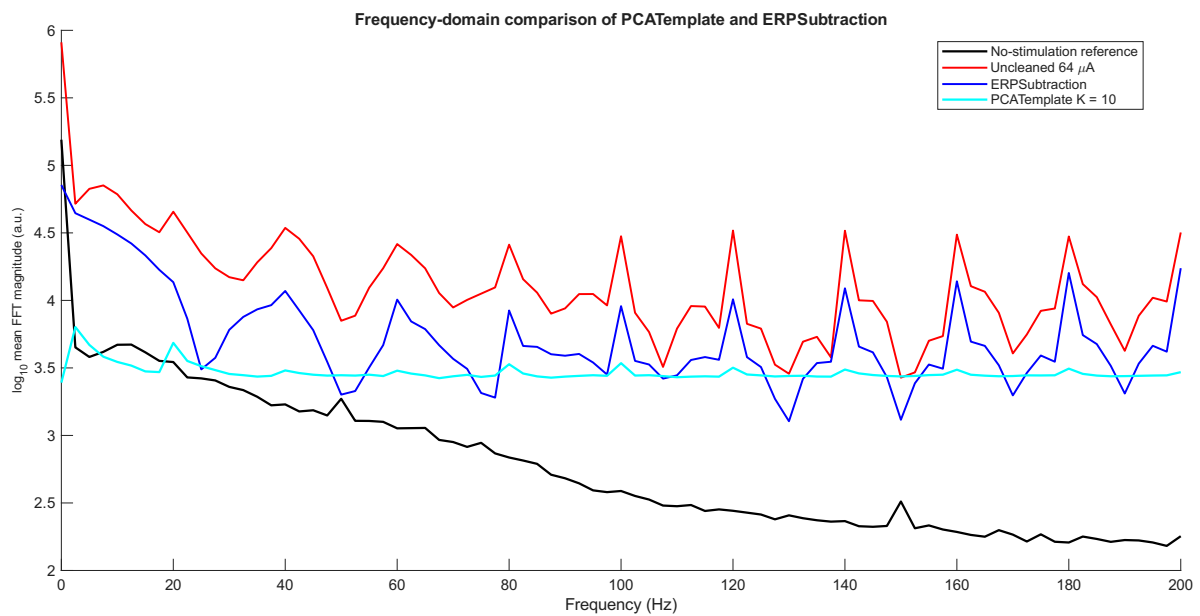


Fig. 4.3 Frequency-domain comparison of PCATemplate and ERPSubtraction for an example analyzed electrode. PCATemplate with $K = 10$ reduced the harmonic structure more strongly than ERPSubtraction for this example.

PCA was effective at suppressing harmonic structure, but it was treated as an aggressive baseline because PCA components may contain both artifact and neural activity.

4.4 SMARTALite Results

SMARTALite methods were evaluated after ERP and PCA baselines. Single- K nearest-neighbor methods and ensemble versions were compared. SMARTALite ensemble K3K5 was initially selected as a lightweight adaptive method because it improved total FFT error compared with raw high-stimulation data and remained interpretable through local pulse-wise templates.

However, after enabling optimal shrinkage for SMARTAFull, SMARTALite was no longer the final selected method. It remained an important intermediate method and lightweight comparison.

This comparison was retained because SMARTALite represents a computationally simpler SMARTA-inspired method, whereas SMARTAFull adds optimal-shrinkage denoising for improved similarity estimation.

4.5 Representative SMARTAFull FFT Example

After resolving the optimal-shrinkage dependency, SMARTAFull was evaluated on a representative non-stimulation electrode. Electrode 7 was used because it showed clear stimulation-related harmonic structure and was part of the selected 31-electrode validation set.

For this representative example, SMARTAFull with $K = 3$ and a 100 Hz high-pass similarity filter produced the lowest total FFT error and strongest harmonic suppression. The raw total FFT error was 12.2846, ERPSubtraction gave 9.1427, PCATemplate K10 gave 7.7298, and SMARTAFull gave 5.9588. The raw harmonic error was 5.9011, while SMARTAFull reduced the harmonic error to 1.8834, corresponding to 68.08% harmonic suppression.

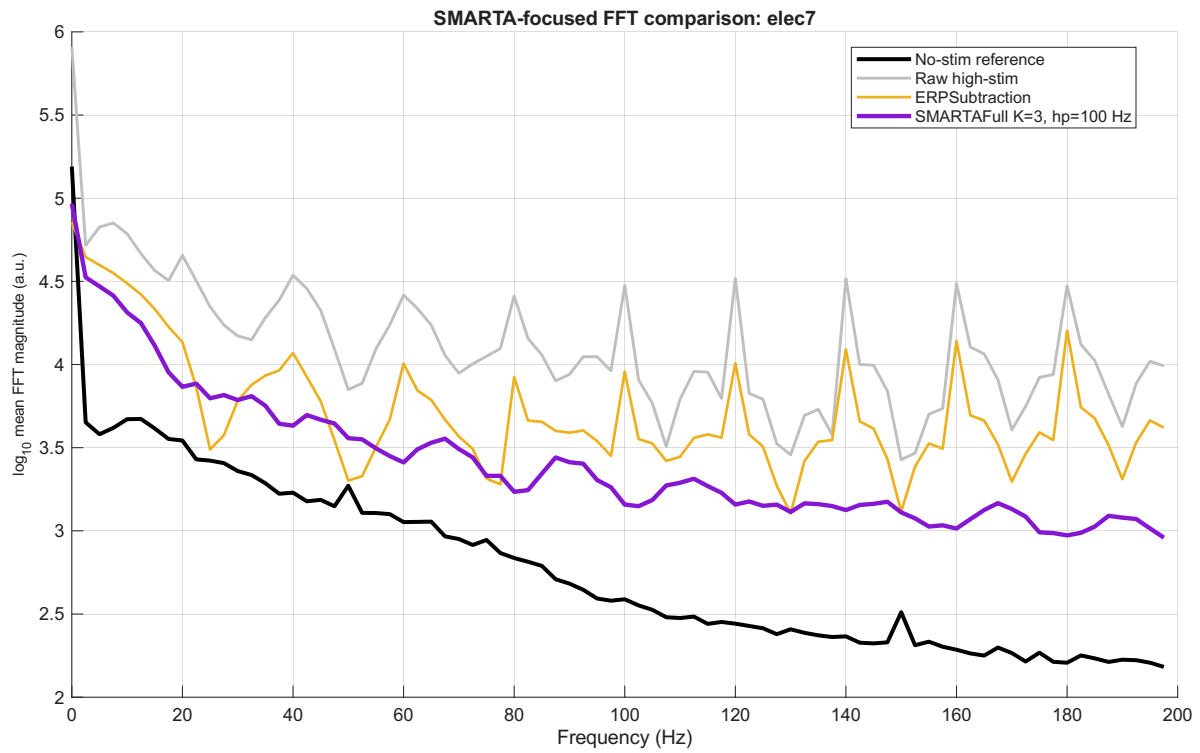


Fig. 4.4 Representative FFT comparison for electrode 7 after optimal-shrinkage SMARTAFull cleaning. The raw high-stimulation signal shows clear stimulation-related harmonic structure. ERPSubtraction reduces part of this structure, while SMARTAFull with $K = 3$ and a 100 Hz high-pass similarity filter gives the lowest FFT error and strongest harmonic reduction in this representative example.

4.6 Pulse-Triggered Residual Artifact

A pulse-triggered residual analysis was performed on electrode 7 to directly evaluate pulse-locked artifact remaining after cleaning. Segments were aligned to each stimulation pulse and averaged. The immediate pulse artifact was measured over 0–6 ms after the pulse, and post-pulse ringing was measured over 6–20 ms.

Table 4.2 Pulse-triggered residual artifact metrics for electrode 7. Lower values indicate lower residual pulse-locked artifact.

Method	Pulse RMS 0–6 ms	Ringing RMS 6–20 ms	Overall RMS
Raw high-stim	334.86	459.15	381.47
ERPSubtraction	70.26	70.26	62.99
PCATemplate K10	66.49	70.26	62.03
SMARTALite ensemble K3K5	66.20	72.43	122.06
SMARTAFull K3 hp100	41.64	53.19	45.39

SMARTAFull produced the lowest immediate pulse artifact, lowest post-pulse ringing, and lowest overall pulse-triggered residual. This supports SMARTAFull as the strongest method for reducing pulse-locked artifact in the representative electrode.

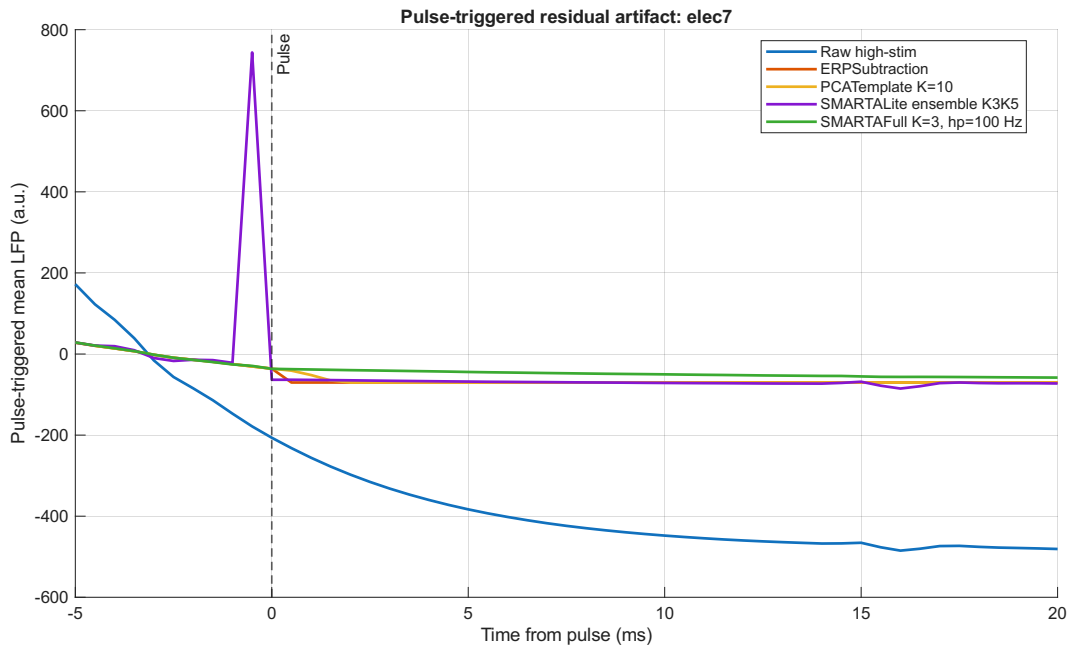


Fig. 4.5 Pulse-triggered residual artifact for electrode 7. The traces were aligned to stimulation pulses and averaged. SMARTAFull produced the lowest residual artifact amplitude and the lowest post-pulse ringing among the tested methods.

4.7 Final Across-Electrode Validation Including SMARTAFull

The final validation was performed across the 31 selected non-stimulating V1 recording electrodes. The methods compared were RawHighStim, ERPSubtraction, PCATemplate K10, SMARTALite ensemble K3K5, and SMARTAFull K3 hp100. The SMARTAFull implementation used optimal shrinkage successfully in the final validation, with meanUsedShrinkageHigh equal to 1.

Table 4.3 Final real-data validation summary across 31 selected V1 electrodes. Lower values are better for error metrics and balanced score.

Method	Total FFT error	Harmonic excess	No-stim FFT distortion	Balanced score
Raw high-stim	13.539	6.1869	–	–
ERPSubtraction	9.8437	4.9232	0.4561	0.4353
PCATemplate K10	11.1580	4.1634	3.0362	0.7162
SMARTALite ensemble K3K5	9.8192	5.0080	3.0571	0.6406
SMARTAFull K3 hp100	7.4393	2.3616	2.3832	0.2642

SMARTAFull achieved the lowest mean total FFT error, reducing the error from 13.539 in the raw high-stimulation condition to 7.4393. It also achieved the lowest mean harmonic excess error and the lowest balanced score. ERPSubtraction remained the most conservative method in the no-stimulation negative-control analysis, as shown by the lowest no-stimulation FFT distortion.

Table 4.4 Best-method counts across 31 electrodes for the main final validation metrics.

Metric	ERP	PCA K10	SMARTALite K3K5	SMARTAFull K3 hp100
Total FFT error	0	0	1	30
Harmonic excess error	0	0	0	31
Harmonic absolute error	0	0	0	31
Below-reference error	24	0	0	7
Trial-STD normalized error	0	22	7	2
No-stim normalized RMS distortion	30	0	0	1
No-stim FFT distortion	31	0	0	0
Balanced score	1	0	0	30

The best-method counts show that SMARTAFull was the most consistent method for artifact suppression. It was best on 30 of 31 electrodes for total FFT error, 31 of 31 electrodes for harmonic excess error, and 30 of 31 electrodes for balanced score. ERPSubtraction was best for no-stimulation distortion, confirming that it was the most conservative method. PCATemplate K10 was strongest for trial-variability preservation on many electrodes, but it did not provide the best overall artifact

suppression or balanced score.

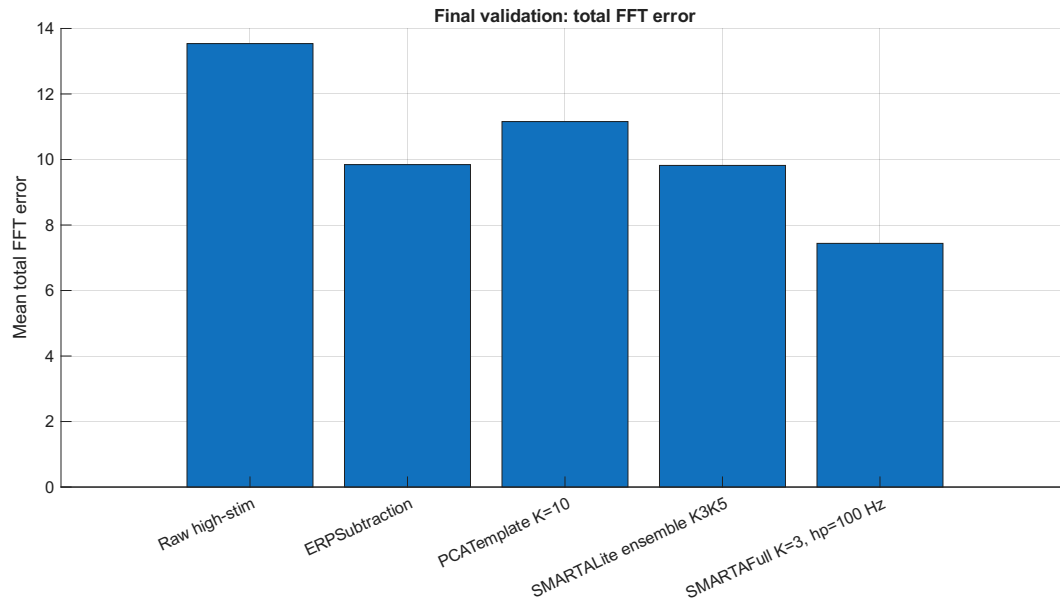


Fig. 4.6 Mean total FFT error across 31 selected V1 electrodes. SMARTAFull with $K = 3$ and a 100 Hz high-pass similarity filter produced the lowest mean total FFT error.

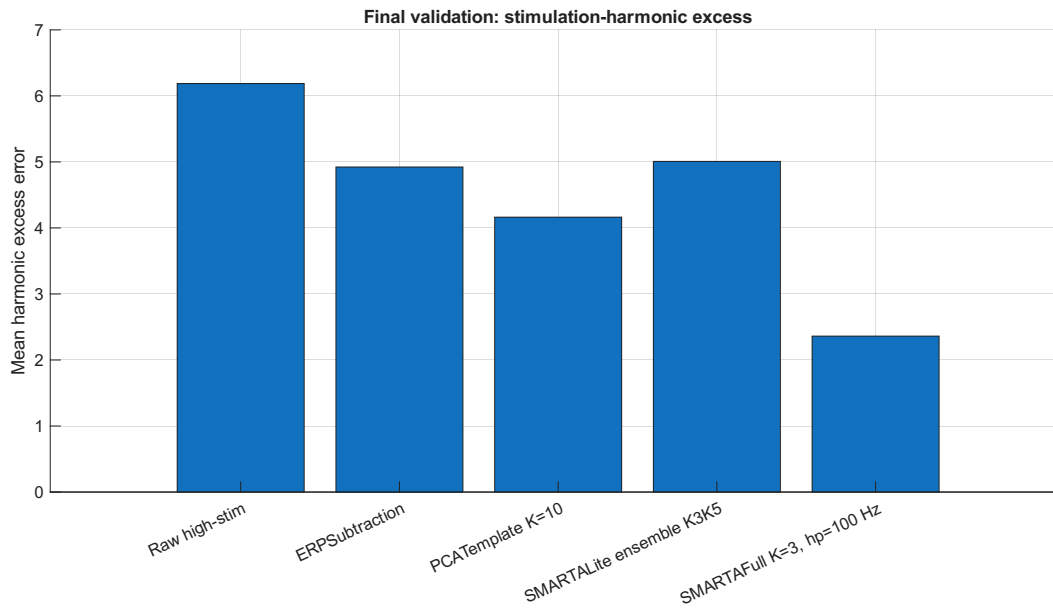


Fig. 4.7 Mean stimulation-harmonic excess error across 31 selected V1 electrodes. SMARTAFull produced the strongest reduction in stimulation-related harmonic excess.

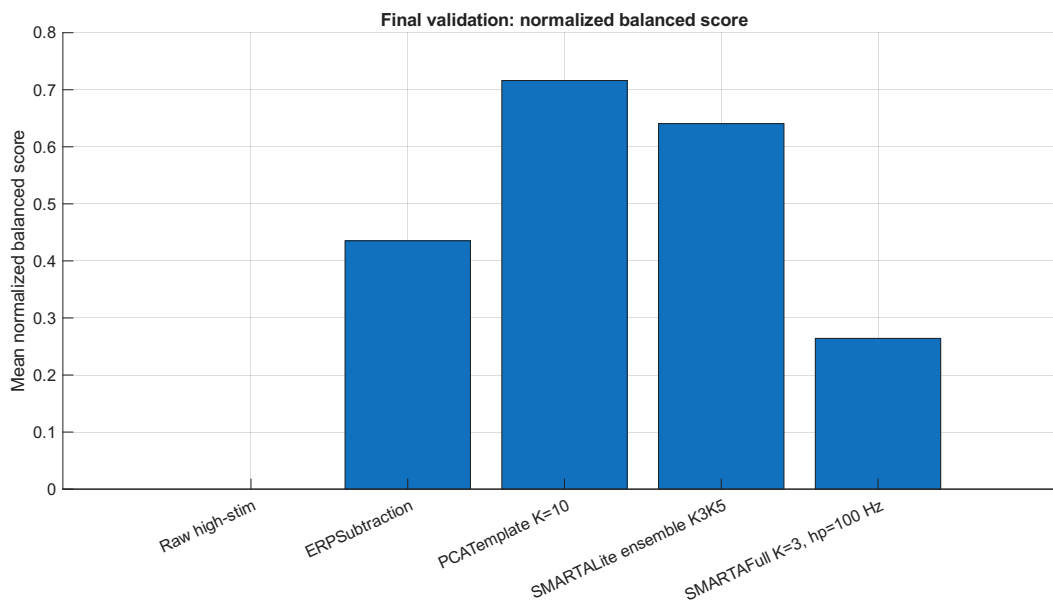


Fig. 4.8 Mean normalized balanced score across 31 selected V1 electrodes. SMARTAFull achieved the lowest balanced score, indicating the best overall trade-off among the tested methods.

4.8 Semi-Synthetic Validation

Semi-synthetic validation was used to evaluate recovery when the clean reference was known. No-stimulation trials were used as clean signals, stimulation artifact estimates were added, and artifact-removal methods were applied to recover the clean signal.

The earlier semi-synthetic validation was performed before SMARTAFull with optimal shrinkage was added to the final comparison. In that validation, PCATemplate K10 gave the strongest known-clean recovery. This result is useful because it shows that PCA can be powerful when the clean target is known. However, PCA was still interpreted cautiously because it can be aggressive and may remove time-locked neural activity in real stimulation data.

After SMARTAFull was enabled, the final real-data validation was repeated and showed that SMARTAFull provided the strongest artifact reduction and best balanced score across electrodes. A future extension would be to rerun the semi-synthetic validation with SMARTAFull included directly.

4.9 Discussion

The results show that different methods have different strengths. ERPSubtraction is simple and conservative. It produces the lowest no-stimulation distortion and therefore provides an important baseline. However, ERPSubtraction leaves residual stimulation-related spectral structure and is not the strongest artifact-suppression method.

PCATemplate K10 reduces stimulation harmonic structure and preserves trial variability well in some metrics, but it can be aggressive. Since PCA removes dominant variance components, these components may contain both artifact and neural activity. Therefore, a low spectral error alone is not sufficient to conclude that PCA is physiologically safest.

SMARTALite ensemble K3K5 provided a lightweight adaptive method and improved on the raw high-stimulation signal. However, after optimal shrinkage was enabled, SMARTAFull clearly outperformed SMARTALite in the final validation. SMARTAFull achieved the lowest total FFT error, lowest harmonic excess error, and lowest balanced score across the 31 selected electrodes.

The final selected method is therefore SMARTAFull with $K = 3$ and a 100 Hz high-pass similarity filter. This method provided the strongest artifact reduction while maintaining a favorable

balanced score. ERPSubtraction remains an important conservative baseline, and PCA remains a useful but more aggressive comparison method.

Because the true neural response during ICMS is not directly observable in real high-stimulation trials, these results should be interpreted as evidence of improved artifact suppression and preservation according to the chosen validation metrics, rather than as direct proof of exact neural signal recovery.

4.10 Summary

The experimental validation showed that ERP-based methods provide interpretable baselines, PCA-based methods strongly suppress harmonic artifacts but may be aggressive, and SMARTA-based methods provide adaptive local artifact estimation. After enabling optimal shrinkage, SMARTAFull K3 hp100 became the final selected method. It achieved the best total FFT error, harmonic artifact error, and balanced score across 31 selected V1 electrodes.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This project addressed the problem of stimulation artifact removal in ICMS-evoked LFP recordings. The main challenge was to reduce large pulse-locked electrical artifacts while preserving the underlying neural signal. Several methods were implemented and compared, including ERP-based template subtraction, PCA-based subtraction, SMARTALite, and SMARTAFull.

ERPSubtraction was found to be the most conservative baseline because it produced the smallest no-stimulation distortion. PCA-based subtraction reduced stimulation-related harmonic components but was treated cautiously because component removal can be aggressive and may remove time-locked neural activity. SMARTALite provided a lightweight adaptive nearest-neighbor template method and served as an important intermediate approach.

The final selected method was SMARTAFull with $K = 3$ and a 100 Hz high-pass similarity filter. After enabling optimal-shrinkage denoising, SMARTAFull achieved the lowest mean total FFT error, lowest stimulation-harmonic excess error, and lowest balanced score across the 31 selected non-stimulating V1 recording electrodes. It was best on 30 of 31 electrodes for total FFT error, 31 of 31 electrodes for harmonic error, and 30 of 31 electrodes for balanced score. Therefore, SMARTAFull provided the strongest overall artifact-removal performance for this dataset.

5.2 Future Work

Future work can extend this project in several directions. First, the semi-synthetic validation should be rerun with SMARTAFull included, so that known-clean recovery can be evaluated directly for the final selected method. Second, SMARTAFull parameters such as K , high-pass cutoff frequency, and shrinkage settings can be optimized using cross-validation across electrodes. Third, additional preservation metrics can be developed to test whether stimulation-evoked neural responses are retained after cleaning. Finally, the method can be tested on other ICMS protocols, including frequency-varying or pulse-number-varying protocols, to evaluate robustness across stimulation settings.

Appendix A

Implementation Notes

The final SMARTAFull validation required the helper function `createPseudoNoise.m` for the optimal-shrinkage routine. The active MATLAB path used the project versions of `SMARTAFull.m` and `optimal_shrinkage_color_fast.m`, with `createPseudoNoise.m` added from the original SMARTA repository. The final validation confirmed that optimal shrinkage was used successfully by reporting `meanUsedShrinkageHigh = 1` for SMARTAFull.

The preliminary GUI codebase and common analysis utilities used for accessing the GRF protocol data were developed in the NeuroOscillations Laboratory. These routines were used for data loading, condition selection, and consistency with the existing ICMS analysis framework. The artifact-removal scripts, SMARTA-based comparisons, final validation metrics, and thesis-specific figures were developed around this existing framework.

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